Student Name:
Student Exam number:
GitHub User:
GitHub Repo:

Abstract

Describe the problem that the report addresses in context of the game domain.
Outline how the developed game addresses the requirement – its key characteristics and fundamental principles (establishing a solution).

Introduction

The introduction must describe the game.

Requirements

Describe the component-based game in terms **of interface contracts**, functional and non-functional requirements.
The game must include **Player**, **Enemy**, Asteroids, **Weapon/Bullet** and Rendering/Drawing (Core) components.
The Player, Enemy and Weapon components must implement service provided interfaces that allow the components to be updated and removed without recompilation.

Analysis

Analysis describes only **what** the system should do and not **how** it is done.
In analysis, you can come up with a rough draft of the interfaces and the entities of the game.
Furthermore, you should document use cases/gameplay, the object model using a UML class diagram and the communication between components with sequence diagrams.

Design

The design describes **what** the structure of the system should be to fulfill the requirements.
Document the architecture and abstractions of the system.
Design develops those abstractions into realizable components.
Describe and sketch the **component models** of the game using a UML component diagram, see [UML] .
The component contracts in the system must be described in terms of pre- and post-conditions.
Furthermore, the different elements of the game and how they are connected must be described.

Implementation

In implementation, you document the implementation (code) of the components from design.
Describe the details of how the components are registered and accessed.
How are reliable dependencies and strong encapsulation enforced in your project?
What component models are applied and where in the source code?
Provide a descriptive explanation of each element in the implementation and provide arguments for your choices.

You should describe how you register components and how you access them.
Note, you are allowed to reuse the game logic, but you must implement the **[GameLab], [JavaLab]**, **[JPMSLabs]**, **[SpringLab], [TestLab]** and **[MicroServiceLab]** labs based on your own GitHub repositories.

Test

Describe how experimental validation was performed through deployment of the game on top of the component container in a real setting.
Test the system's software-abilities such as dynamic updates using integration and unit test.

Discussion

Discuss how well the game solved the identified essential problems (module updates etc.).
To which extent did your design meet the requirements?

Conclusion

First summarize the report.
Remember that you are summarizing the report for a reader that has read the introduction and the body of the report already and has a strong sense of key concepts and applied technologies.
Explain the potential impacts of your system in relation to the main issue.
Direct future work directions related to the main issue.
However, this should not be seen as an opportunity to develop new ideas in significant detail and should be clearly linked to the work described in your report.

References

Please use IEEE citation style,